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Electrodynamics and Hemodynamics of the Human Organism: A Comprehensive Synthesis of Biological Interaction Mechanisms

1. Introduction: The Convergence of Biophysics and Electromagnetics

The study of the human body's interaction with external fields is a discipline that bridges the discrete worlds of fluid dynamics, cellular physiology, and electromagnetic theory. The inquiry initiates with the fundamental understanding of the cardiovascular system—specifically the hemodynamic shear stress acting upon the endothelium—as a baseline for biological mechanotransduction, and extends into the complex domain of exogenous electromagnetic fields (EMF) ranging from extremely low frequencies (ELF) to the millimeter-wave bands of modern telecommunications. This report provides an exhaustive, expert-level synthesis of these interactions, integrating the seminal work on endothelial shear stress ¹ with the vast body of dielectric and dosimetric research defined by the Gabriel datasets ² and contemporary investigations into 5G and transient pulse phenomena.

The human organism is not a passive dielectric object; it is a dynamic volume of conductive fluids, polarized membranes, and electrochemical signaling pathways. To understand "frequency human body interaction," one must first appreciate the baseline physiological state—a state defined by the pulsatile flow of blood, the orientation of endothelial cells under shear stress, and the delicate ionic balances across voltage-gated channels. When external electromagnetic forces are superimposed upon this biological substrate, the outcomes are governed by a complex matrix of variables: frequency, modulation, tissue permittivity, conductivity, and the thermodynamic capacity of the vascular system to dissipate generated heat.

This analysis is structured to traverse the full spectrum of interaction. We begin by establishing the biophysical properties of human tissue, utilizing the parametric models of Gabriel et al. (1996) as the ground truth. We then integrate the specific pathophysiology of hemodynamic shear stress, examining how electromagnetic fields might couple with the cardiovascular system through thermal or magnetohydrodynamic mechanisms. Subsequently, we dissect the mechanisms of interaction across the frequency spectrum—from the induced

currents of 50 Hz power lines to the skin-limited absorption of 60 GHz millimeter waves—and conclude with an analysis of transient phenomena like Brillouin precursors and ultrashort pulses, which challenge conventional dosimetric assumptions.

1.1 The Biophysical Substrate: Blood, Tissue, and Dielectrics

At the macroscopic level, the human body is a lossy dielectric conductor. Its response to an electromagnetic field is dictated by its complex permittivity ($\hat{\epsilon}$), a frequency-dependent parameter that describes both the ability of the tissue to store electrical potential energy (polarization) and its tendency to dissipate that energy as heat (conduction).³

The complex relative permittivity is defined as:

$$\hat{\epsilon}(\omega) = \epsilon'(\omega) - j\epsilon''(\omega) = \epsilon'(\omega) - j\frac{\sigma(\omega)}{\omega\epsilon_0}$$

Where ϵ' is the real part (dielectric constant), ϵ'' is the imaginary part (loss factor), σ is the total conductivity (ionic + dielectric loss), ω is the angular frequency, and ϵ_0 is the vacuum permittivity.

The fundamental challenge in bioelectromagnetics is that $\hat{\epsilon}$ is not constant. It varies largely with tissue type (high-water vs. low-water content) and frequency. The research established by C. Gabriel and colleagues identifies three primary dispersion regions in biological tissues ²:

1. **α -dispersion (10 Hz – 10 kHz):** Dominated by the diffusion of counter-ions along the charged surfaces of cell membranes. This results in colossal permittivity values (often $>10^6$) in tissues like muscle and blood, as the cell membranes act as large capacitors charging and discharging with the slow oscillation of the field.
2. **β -dispersion (10 kHz – 100 MHz):** As frequency increases, the ions can no longer follow the field, and the cellular membrane capacitance is shorted out. The field begins to penetrate the intracellular medium, and the charging of organelle membranes becomes the dominant polarization mechanism.
3. **γ -dispersion (> 100 MHz – 100 GHz):** This is the regime of water. The interaction is dominated by the dipolar rotation of free water molecules. Since soft tissues are approximately 70-80% water, this dispersion defines the absorption characteristics for microwave and millimeter-wave technologies.

Understanding these regimes is critical to the user's query regarding "frequency human body interaction." A 60 Hz field sees the body as a collection of insulating membranes and conductive pathways, inducing currents primarily in the extracellular space. A 28 GHz field sees the body as a semi-infinite slab of lossy water, depositing energy almost exclusively in

the superficial layers of the skin.⁶

2. Hemodynamics and Endothelial Shear Stress: The Physiological Baseline

Before assessing the impact of external fields, we must characterize the endogenous mechanical environment of the cardiovascular system, as referenced in the user's specific request regarding PMC2851404.¹ The endothelium, the inner lining of blood vessels, is a mechanotransducer that regulates vascular health in response to hemodynamic shear stress—the frictional force exerted by flowing blood.

2.1 The Role of Shear Stress in Pathophysiology

Shear stress (τ) is proportional to the viscosity of blood (μ) and the gradient of velocity (du/dy) at the vessel wall:

$$\tau = \mu \frac{\partial u}{\partial y}$$

Physiological shear stress maintains endothelial quiescence, promoting the release of vasodilators like nitric oxide (NO) and inhibiting inflammatory pathways. Conversely, disturbed flow (low or oscillatory shear stress), which occurs at arterial bifurcations, activates pro-inflammatory genes (e.g., NF- κ B), making these sites prone to atherosclerosis.¹

2.2 Electromagnetic Coupling to Hemodynamics

The interaction between EMF and this hemodynamic environment is multifaceted, involving both thermal and non-thermal fluid dynamic coupling:

2.2.1 Thermal Vasodilation and Flow Modification

The primary mechanism by which RF and microwave fields interact with hemodynamics is thermal. Blood is a high-water-content tissue with high conductivity (~ 1.0 S/m at 900 MHz).⁸ When the body is exposed to RF energy (e.g., from a mobile phone or MRI), energy is dissipated in the blood and surrounding tissues. The body's thermoregulatory response is to increase local blood perfusion to carry heat away from the "hot spot."

This increase in flow velocity alters the shear stress profile at the vessel wall. While moderate increases in shear stress are generally atheroprotective, rapid or extreme thermal gradients can induce turbulent flow or alter the viscosity of blood (which decreases with temperature), potentially destabilizing the local hemodynamic equilibrium described by Davies.¹

2.2.2 Magnetohydrodynamics (MHD)

In the presence of strong static magnetic fields (such as those in MRI scanners, > 1.5 T), a non-thermal interaction occurs. Blood, being a conductive fluid containing electrolytes (Na^+ , Cl^- , K^+), moving through a magnetic field (B) generates an induced

voltage (Hall effect) and a Lorentz force ($\mathbf{F}$) perpendicular to both the velocity ($\mathbf{v}$) and the field:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$$

This force can retard blood flow (the magnetohydrodynamic effect). While generally negligible at environmental exposure levels (microtesla range), at high field strengths, this effect can distort the blood velocity profile, theoretically increasing the pressure required to pump blood (hemodynamic drag) and altering the shear stress vectors acting on the endothelium. This represents a direct physical intersection between the user's "frequency interaction" query and the "hemodynamic shear stress" literature.¹⁰

3. The Gabriel Dielectric Data: Mapping the Body's Impedance

To model these interactions accurately, we rely on the foundational measurements by Gabriel et al. (1996), which provided the dielectric properties for over 30 tissue types.² These data sets are essential for finite-difference time-domain (FDTD) simulations used in modern dosimetry.

3.1 Tissue-Specific Dielectric Profiles

The body is heterogeneous. The interaction of a wave with the "human body" is actually an interaction with a stratified medium of skin, fat, muscle, and bone, each with distinct electrical properties.

- **Muscle and Heart:** These are high-water-content tissues. At 50 Hz, muscle has an extremely high relative permittivity ($\epsilon_r \approx 10^7$) due to membrane capacitance, but relatively low conductivity ($\sigma \approx 0.2$ S/m). As frequency rises to 900 MHz (cellular range), permittivity drops to ~ 55 , and conductivity rises to ~ 1 S/m. This makes muscle a primary absorber of RF energy.⁷
- **Adipose Tissue (Fat):** Fat has low water content. Its permittivity (~ 5 - 10 at RF) and conductivity are an order of magnitude lower than muscle. This creates an impedance mismatch at the fat-muscle interface. When an EM wave traverses this boundary, reflections occur, leading to standing waves that can create localized "hot spots" of SAR (Specific Absorption Rate) in the muscle layer immediately adjacent to the fat.¹³
- **Bone:** Cortical bone is highly resistive and acts as an insulator at low frequencies. However, cancellous bone (containing marrow and blood) is more conductive. At microwave frequencies, bone becomes relatively transparent compared to muscle, allowing energy to propagate into the skull and affect the brain.²

3.2 Frequency Dependence and the Cole-Cole Model

Gabriel et al. synthesized their data into a 4-term Cole-Cole equation to predict dielectric properties at any frequency:

$$\hat{\epsilon}(\omega) = \epsilon_{\infty} + \sum_{n=1}^4 \frac{\Delta\epsilon_n}{1 + (j\omega\tau_n)^{(1-\alpha_n)}} + \frac{\sigma}{j\omega\epsilon_0}$$

This equation accounts for the multiple relaxation processes (α , β , γ , δ) occurring simultaneously. For example, the measurement of **Human Skin (wet)** shows a transition from a highly resistive barrier at ELF (protecting internal organs from electric fields) to a highly absorptive medium at 60 GHz (where $\sigma > 30$ S/m), effectively trapping energy at the surface.¹⁵

4. Extremely Low Frequency (ELF) Interactions (0 – 300 Hz)

Moving through the frequency spectrum, the first major regime is the Extremely Low Frequency band, dominated by 50/60 Hz power transmission. Here, the wavelength (~ 5000 km) is vastly larger than the human body. The body does not act as an antenna; instead, the interaction is governed by quasi-static induction.

4.1 Mechanisms of Action: Induction and Stimulation

In this regime, the primary metric of concern is the **induced electric field** (E_{int}) within the tissue. An external oscillating magnetic field (B) passes through the body largely unperturbed (since tissue magnetic permeability $\mu \approx \mu_0$). However, according to Faraday's Law of Induction, this changing magnetic flux induces circulating eddy currents within the conductive tissues of the body.¹⁷

$$J = \sigma E_{\text{int}} \approx \sigma (\pi f r B)$$

Where J is the current density, f is frequency, and r is the loop radius (e.g., the radius of the torso).

The biological hazard at high intensities is **electrostimulation**. If the induced field exceeds the threshold for depolarization of nerve membranes, it can trigger action potentials.

- **Peripheral Nerve Stimulation (PNS):** Can cause tingling or muscle twitching.
- **Central Nervous System (CNS) Effects:** The retina is the most sensitive tissue. Induced fields as low as 50-100 mV/m can trigger **magnetophosphenes**—the perception of flickering lights caused by the stimulation of retinal neurons.¹⁸

4.2 The "Voltage-Gated Ion Channel" (VGIC) Hypothesis

Beyond acute stimulation, a critical area of "Deep Research" is the mechanism for effects at

sub-threshold levels. Panagopoulos et al.¹⁹ have proposed a biophysical mechanism where ELF magnetic fields exert forces on the free ions (e.g., Ca^{2+} , K^{+}) that control the gating of voltage-gated ion channels.

- **Mechanism:** Coherent, polarized ELF fields may induce an irregular forced oscillation of these ions. Even if the thermal energy is high, the coherent nature of the external field could theoretically disrupt the delicate gating mechanism, causing channels to open or close inappropriately.
- **Consequence:** A dysfunction in Ca^{2+} channels can lead to intracellular calcium overload. This, in turn, can trigger the overproduction of Reactive Oxygen Species (ROS) by mitochondria, leading to oxidative stress. This pathway links ELF exposure to DNA damage and genotoxicity without requiring the energy to break chemical bonds directly.¹⁹

4.3 Epidemiological Context

The IARC classifies ELF magnetic fields as "possibly carcinogenic" (Group 2B), based largely on consistent epidemiological associations between high-voltage power lines and childhood leukemia (at levels $>0.4 \mu\text{T}$). While animal studies have struggled to replicate this⁴⁹, the VGIC/ROS mechanism provides a plausible, albeit debated, biological rationale for how such weak fields could influence cellular pathology over chronic durations.¹⁷

5. Radiofrequency (RF) Interactions (100 kHz – 6 GHz)

As frequencies rise into the MHz and GHz range (FM radio, TV, WiFi, 4G), the interaction physics shifts from induction to **resonance** and **dielectric heating**.

5.1 Whole-Body Resonance and SAR

The human body acts as a resonant antenna in the RF band. For a grounded adult human, the resonant frequency is approximately 30-40 MHz; for an isolated human (e.g., standing on a wooden floor), it is 70-100 MHz. At resonance, the body absorbs RF energy with maximum efficiency.²²

To quantify this absorption, we use the Specific Absorption Rate (SAR):

$$\text{SAR} = \frac{\sigma |E|^2}{\rho} \quad \text{W/kg}$$

Where $|E|$ is the RMS internal electric field and ρ is tissue density.

SAR is the regulatory benchmark for this band. It is a measure of the rate of energy deposition. If the global SAR exceeds the body's metabolic cooling capacity (roughly 1-4 W/kg for extended periods), the core body temperature will rise.

5.2 Thermal Effects and Hemodynamic Response

The connection to hemodynamic shear stress¹ is most direct here. An increase in local tissue

temperature triggers a potent vasodilatory response. Capillaries and arterioles dilate to increase blood flow and convect heat away.

- **Physiological Cascade:**

1. RF Energy deposition $\rightarrow$ Local Temperature Rise.
2. Thermoreceptors trigger autonomic response.
3. Endothelial cells release NO; smooth muscle relaxes.
4. **Flow Velocity Increases:** This changes the shear stress profile (τ) on the endothelium. While acute, moderate shear stress increase is physiologic, chronic thermal stress or extreme gradients (as seen in high-power occupational exposure) could induce pathological flow patterns or thermal damage to the endothelial lining itself.²³

5.3 The Non-Thermal RF Controversy

While ICNIRP guidelines²⁵ are based on preventing thermal effects (threshold of 1°C rise), substantial literature argues for non-thermal effects in the RF band:

- **Blood-Brain Barrier (BBB) Permeability:** Studies have reported that exposure to GSM-modulated fields (900 MHz) can increase the permeability of the BBB, allowing albumin and other solutes to leak into brain tissue. This effect appears to be non-linear and may depend on the specific modulation (pulsing) of the signal, although replication has been inconsistent.²³
- **Genotoxicity and ROS:** Similar to the ELF mechanism, RF exposure has been linked to increased oxidative stress in neurons and sperm cells. Snippet²⁸ notes that high-power or long-term microwave exposure caused alterations in spermatogenesis (reduced motility, abnormal morphology), likely mediated by ROS generation rather than bulk heating.

6. Millimeter Waves and the 5G Era (> 6 GHz)

The deployment of 5G networks utilizing the FR2 band (24 GHz – 100 GHz) represents a fundamental shift in bio-interaction due to the physics of **skin depth**.

6.1 The Skin as the Principal Target

At frequencies above 6 GHz, the penetration depth (δ) of electromagnetic waves decreases drastically.

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}$$

At 28 GHz (a common 5G band), the penetration depth in skin is less than 1 mm. At 60 GHz, it is roughly 0.5 mm.⁵

- **Implication:** The skin and the cornea of the eye become the sole absorbers of incident

energy. The internal organs (brain, heart, liver) are effectively shielded by the body's own surface tissues.

- **Dosimetry Shift:** Consequently, SAR (a volumetric metric) becomes inappropriate. Safety standards have transitioned to **Absorbed Power Density (S_{abs})** or **Incident Power Density (S_{inc})**, measured in W/m^2 .²⁹

6.2 Biological Effects of Millimeter Waves (MMW)

1. **Cutaneous Heating and Nociception:** The skin is densely innervated with free nerve endings (nociceptors) and thermoreceptors. MMWs can heat these endings rapidly. This principle is utilized in Active Denial Systems (95 GHz) to create an intolerable burning sensation for crowd control. In the context of 5G, the concern is whether lower-level, chronic heating of these nerves could trigger systemic reflexes or altered pain thresholds.⁶
2. **Systemic Signaling:** There is evidence that MMW interaction with skin can have distal effects. Snippet³¹ discusses how MMWs may modulate the release of endogenous opioids or cytokines from skin cells, potentially affecting the immune system or pain perception systemically. This suggests the skin acts as a transducer, converting surface EM signals into systemic biochemical signals.
3. **Bacterial and Cellular Effects:** Some studies³² suggest MMWs can alter the growth rates of bacteria (both enhancement and inhibition observed) and affect antibiotic resistance. The mechanism is hypothesized to be changes in membrane transport or water structuring at the cellular interface.

7. Transient Phenomena: Brillouin Precursors and Ultrashort Pulses

The most advanced and technically complex area of "Deep Research" in this domain involves the time-domain behavior of signals, specifically those with ultra-wide bandwidths or extremely fast rise times.

7.1 Brillouin Precursors: Penetrating the Impenetrable?

Conventional theory dictates that high-frequency signals attenuate rapidly in lossy media like tissue (the skin depth limit discussed above). However, this assumes a continuous wave or a slowly varying envelope.

Oughstun's Theory: When a pulse with an extremely rapid rise time enters a dispersive medium (where refractive index varies with frequency), the wave packet distorts. A transient component known as the Brillouin precursor can form.

- **Characteristics:** The Brillouin precursor travels at a velocity different from the main signal velocity and, crucially, decays *algebraically* (e.g., as $z^{-1/2}$) rather than *exponentially* (as $e^{-\alpha z}$).
- **Implication:** Theoretically, this allows a fraction of the pulse energy to penetrate much

deeper into the body than the carrier frequency would suggest. Snippets³⁴ highlight that this could enable UWB (Ultra-Wideband) radar or high-speed data pulses to reach deep tissues (brain, heart) that should otherwise be shielded.

- **Health Risk:** If 5G or 6G signals generate significant precursors, the safety assessments based on "skin depth" shielding might be underestimates. The precursor could deposit energy in deep anatomical structures, potentially interacting with deep-seated neural circuits or the myocardium.³⁷

7.2 Ultrashort Pulses (nsPEF) and Intracellular Electroporation

Pulse width is a critical variable. Conventional electroporation (using microsecond/millisecond pulses) charges the outer cell membrane, causing pore formation. However, **nanosecond pulsed electric fields (nsPEFs)** are too fast to charge the outer membrane fully.

- **Mechanism:** Instead of affecting the plasma membrane, the electric field penetrates the cell interior and charges the membranes of organelles (mitochondria, endoplasmic reticulum, nucleus).
- **Effect:** This can trigger calcium release from the ER (mimicking a physiological signal) or induce apoptosis (programmed cell death) by permeabilizing mitochondria—all without destroying the cell. This phenomenon is being explored for cancer therapy (ablating tumors without thermal necrosis) but also represents a potential mechanism for bio-effects from UWB noise or radar pulses.³⁸

8. Dosimetry and Regulatory Frameworks

The translation of this complex physics into safety standards is a point of global contention.

8.1 ICNIRP Guidelines (2020)

The International Commission on Non-Ionizing Radiation Protection (ICNIRP) sets the global standard. Their 2020 guidelines are explicitly **thermally based** (for >100 kHz).

- **Philosophy:** ICNIRP maintains that the only substantiated adverse health effects are nerve stimulation (at low frequencies) and thermal damage (at high frequencies).
- **Limits:**
 - **Whole Body SAR:** 0.08 W/kg (General Public).
 - **Local SAR (Head/Torso):** 2.0 W/kg (averaged over 10g of tissue).
 - **Power Density (>6 GHz):** 200 W/m^2 (Occupational), 10–40 W/m^2 (Public).⁴⁰
- **Rationale:** These limits include safety factors (typically 10–50x) below the threshold of 4 W/kg, which is known to raise body temperature by $\sim 1^\circ\text{C}$ in animal models (e.g., the rat study in⁴²).

8.2 The "BioInitiative" and Alternative Views

Contrastingly, groups like the BioInitiative Working Group argue that the thermal model is

insufficient. They cite the vast literature on VGIC dysfunction, ROS production, and DNA damage at non-thermal levels (levels where SAR ≤ 4 W/kg).

- **The Safety Gap:** Critics argue that ICNIRP ignores the "precursor" physics and the "window effects" (where specific frequencies or intensities trigger biological responses not seen at higher or lower levels). They advocate for limits based on biological markers (e.g., oxidative stress) rather than thermal thresholds.⁴³

8.3 Advanced Dosimetry: The Virtual Family

To resolve these disputes, modern dosimetry uses high-resolution anatomical models like the "Virtual Family" (Christ et al., IT'IS Foundation). These are voxelized models of humans (Male, Female, Child) derived from MRI data.

- **Process:** FDTD simulations inject EM waves into these models, using the Gabriel dielectric data for each voxel (brain, bone, blood).
- **Outcome:** This allows researchers to calculate the exact E-field and SAR in specific organs (e.g., "What is the SAR in the hypothalamus of a 5-year-old child using a 900 MHz phone?"). These simulations have revealed that children, with thinner skulls and different dielectric properties, can absorb significantly more energy in the brain cortex than adults.⁴⁶

9. Conclusion

The interaction of electromagnetic frequencies with the human body is a multi-scale phenomenon that defies simple categorization. It is not merely a question of "heating," but a complex interplay of dielectric dispersion, magnetohydrodynamics, ion channel kinetics, and transient wave propagation.

From the hemodynamic perspective initiated by the user's reference 1, we see that EMF can act as a modifier of the vascular environment—altering flow through thermal vasodilation or exerting direct Lorentz forces on the blood stream.

From the dielectric perspective 2, the body is a frequency-dependent filter, transparent to magnetic fields at 50 Hz but opaque to electric fields at 60 GHz.

From the biological perspective, the evidence suggests a duality: a clear, robust thermal mechanism that forms the basis of current safety laws, and a subtle, complex non-thermal landscape involving oxidative stress and ionic disequilibrium that remains the frontier of safety science.

As technology pushes into the millimeter-wave and terahertz domains (6G), the "human body" in the equation changes. We are no longer volumetric absorbers but surface absorbers. The skin becomes the interface for all interaction. The potential for Brillouin precursors and ultrashort pulses to breach this interface represents the most critical biophysical question for the next decade of research.

Table 1: Dielectric Properties of Key Human Tissues Across the Frequency Spectrum

Synthesized from Gabriel et al. (1996) ² and IT'IS Foundation Database.⁴⁸

Tissue	Frequency	Conductivity (σ)	Relative Permittivity (ϵ_r)	Penetration Depth	Dominant Mechanism
Muscle	50 Hz	0.23	2.5×10^7	> 100 m	α -dispersion (Counter-ion diffusion)
	900 MHz	0.94	55.0	~3-4 cm	γ -dispersion (Water rotation)
	24 GHz	20.0*	22.0	< 2 mm	Surface absorption
Blood	50 Hz	0.70	5.0×10^3	> 100 m	Ionic Conduction / MHD Effect
	900 MHz	1.54	61.4	~2-3 cm	Dielectric Loss / Thermal Convection
	24 GHz	~25.0	~20.0	< 1 mm	Rapid Dissipation
Skin (Wet)	50 Hz	0.0002 (Dry)	1100	N/A	Resistive Barrier (Stratum

					Corneum)
	900 MHz	0.87	41.0	~4 cm	Moderate Absorption
	60 GHz	> 35.0	~10.0	~0.5 mm	Total Surface Absorption
Brain (Grey)	50 Hz	0.08	4×10^7	> 100 m	Induced E-Fields / Nerve Stimulation
	900 MHz	0.94	52.7	~3 cm	SAR Hotspots (Standing Waves)
	24 GHz	~15.0	~18.0	Shielded	Protected by Skull/Skin

*Note: High-frequency conductivity values are extrapolated from trends in mmWave dosimetry studies.⁶

Table 2: ICNIRP 2020 vs. FCC Exposure Limits

Comparing the two major regulatory frameworks.²⁵

Metric	ICNIRP 2020 (General Public)	FCC (General Public)	Physiological Basis
Whole Body SAR	0.08 W/kg	0.08 W/kg	Prevention of systemic heat stress ($>1^{\circ}\text{C}$ core rise).
Local SAR	2.0 W/kg (avg. over	1.6 W/kg (avg. over	Prevention of local

(Head/Trunk)	10g)	1g)	tissue damage (e.g., eye, brain). FCC is stricter on peak volume.
Power Density (>6 GHz)	10 - 40 W/m^2	10 W/m^2	Prevention of skin burns/thermal pain.
Averaging Time	30 minutes (Whole body)	30 minutes	Accounts for body's thermal time constant.

This synthesis confirms that while our understanding of the *thermal* physics of interaction is mature, the *non-thermal* and *transient* domains remain areas of active, high-stakes research, particularly as the global electromagnetic environment becomes denser and more spectral diverse.

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